Software Requirements Specification

for

DisasterConnect

A Centralized Disaster-Management and Relief-Coordination Platform

Version 1.0 (Draft)

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1. Introduction

1.1 Purpose

This document specifies the software requirements for Version 1.0 of DisasterConnect, a centralized disaster-management and relief-coordination platform. It describes the functional and non-functional requirements of the system and is intended to serve as the primary reference for design, implementation, and testing of the project. This SRS covers the complete system, including the citizen-facing, volunteer-facing, NGO-facing, authority-facing, and administrator-facing components.

1.2 Document Conventions

Functional requirements are uniquely identified using the format REQ-<module>-<number> (for example, REQ-HELP-1 refers to requirement 1 under the Emergency Help Requests module). Unless otherwise stated, all requirements are of equal priority within their module; where priority differs, it is stated explicitly as High, Medium, or Low. Headings follow a numbered hierarchy consistent with this table of contents.

1.3 Intended Audience and Reading Suggestions

This document is intended for the project supervisor/evaluator, the development team (including the author, as sole developer of this individual project), and any reviewers assessing the system design. Section 2 provides a high-level overview of the product and its users and is recommended as a starting point. Section 3 provides detailed functional requirements organized by module and is the core reference during implementation. Sections 4 and 5 describe interface and non-functional requirements respectively and are most relevant during design and testing.

1.4 Project Scope

DisasterConnect is a web-based platform that connects disaster-affected citizens with emergency services, shelters, NGOs, volunteers, relief resources, and family-safety services. During a disaster (flood, earthquake, cyclone, fire, landslide, or other local emergency), affected citizens can receive alerts, report incidents, request emergency assistance, locate nearby shelters and essential services, and report or search for missing family members. NGOs, volunteers, and emergency authorities use the same platform to view and respond to help requests, coordinate volunteers, manage relief resources, identify shortages, and run verified relief campaigns. An administrator oversees user verification, content moderation, and overall system management.

The goal of the system is to reduce the communication gap between people who need help during a disaster and the organizations or individuals who can provide it, by offering a single, structured platform in place of fragmented information sources such as scattered social-media posts and word of mouth.

1.5 References

- Wiegers, K. E., "Software Requirements Specification Template," Process Impact, 2002 (template basis for this document).
- DisasterConnect — Complete Project Details (internal project brief), consolidated project notes used as the source for this SRS.

2. Overall Description

2.1 Product Perspective

DisasterConnect is a new, self-contained web application; it is not a follow-on to an existing product. It is designed as a layered system with a browser-based frontend, a REST API backend, and a relational database, allowing it to be extended in future with mobile clients or third-party integrations (for example, government disaster-alert feeds or SMS gateways) without changing its core architecture.

2.2 Product Features

At a high level, DisasterConnect performs the following major functions:

- Publish and deliver disaster alerts to affected users.
- Allow citizens to report incidents occurring around them.
- Allow citizens to raise emergency help requests (medical, rescue, food, water, shelter, transportation, etc.).
- Allow citizens to discover nearby shelters and essential services.
- Allow NGOs and volunteers to view, accept, and respond to help requests.
- Allow NGOs to manage relief resources and report/track shortages.
- Allow citizens to report and search for missing family members.
- Allow verified NGOs to run relief campaigns.
- Notify users of relevant status changes and updates.
- Allow an administrator to verify users/NGOs and manage system-wide content and activity.

2.3 User Classes and Characteristics

DisasterConnect supports five user classes, distinguished primarily by their role in the disaster-response process and their privilege level within the system.

User Class	Description
Affected Citizen	General public user; may be under stress during use. Can view alerts, report incidents, request help, find services, and manage missing-person reports.
Volunteer	Registered individual who contributes manpower. Can view and accept nearby help requests and update their status.
NGO / Relief Organization	Verified organization managing larger relief operations, resources, volunteers, and campaigns.
Emergency / Response Authority	Official body that publishes alerts, monitors incidents, and responds to emergency requests.
Administrator	Manages users, NGO verification, disaster information, reported content, and overall system health. Highest privilege level.

2.4 Operating Environment

DisasterConnect is a web-based application accessed through modern desktop and mobile browsers (Chrome, Firefox, Edge, Safari). The backend is expected to run as a REST API service deployable on a standard Linux server or cloud platform, backed by a relational database. The exact technology stack (e.g., Spring Boot backend, React/Angular frontend, PostgreSQL/MySQL database) is to be finalized during design but must support horizontal scaling to accommodate spikes in usage during active disasters.

2.5 Design and Implementation Constraints

- The system must separate presentation, business logic, and data-access layers (layered architecture).
- Location-based features (nearby services, nearby help requests) depend on the availability of geolocation data from the user's device or manually entered addresses.
- NGO verification must be enforced before an NGO account can publish relief campaigns or appear as a trusted responder.
- The system is developed as an individual academic project within a single semester and must therefore prioritize the core modules described in Section 3 over lower-priority extensions.
- Sensitive data (passwords, contact details, missing-person information) must be stored and transmitted securely (see Section 5.3).

2.6 User Documentation

A short in-app help/FAQ section will be provided covering how to report an incident, raise a help request, and use the missing-person module, since users may be operating the system under stress. A brief administrator guide will also be produced covering user/NGO verification and content moderation workflows.

2.7 Assumptions and Dependencies

- Users have access to an internet-connected smartphone or computer during or shortly after a disaster event.
- NGOs and emergency authorities are willing to register and actively monitor the platform for incoming requests.
- Accurate location data (GPS or manually entered) is available for most help requests and incident reports.
- The initial release does not assume integration with external government alert systems or SMS gateways; these are treated as future enhancements.

3. System Features

This section describes each major functional module of DisasterConnect. Each module lists its priority, a representative stimulus/response sequence, and its detailed functional requirements.

3.1 User Management

3.1.1 Description and Priority

Handles registration, login, role assignment, and profile management for all five user classes. Priority: High.

3.1.2 Stimulus/Response Sequences

- User submits registration details ⟶ system creates an account with the selected role and, for NGOs, marks it as pending verification.
- User logs in with valid credentials ⟶ system authenticates and directs the user to the role-appropriate dashboard.

3.1.3 Functional Requirements

REQ-USER-1: The system shall allow a new user to register with name, email, phone number, password, and role (Citizen, Volunteer, NGO, Authority).

REQ-USER-2: The system shall authenticate users via email/phone and password before granting access to protected functionality.

REQ-USER-3: The system shall allow users to view and update their profile information, including location.

REQ-USER-4: The system shall mark NGO accounts as "Pending Verification" until approved by an administrator.

REQ-USER-5: The system shall enforce role-based access so that each user class can only access functionality permitted for its role.

3.2 Disaster Alerts

3.2.1 Description and Priority

Publishes disaster-related alerts (type, area, severity, safety instructions) to users in or near the affected area. Priority: High.

3.2.2 Stimulus/Response Sequences

- Authority/administrator publishes an alert for an area ⟶ system delivers the alert to users associated with or located in that area.
- User opens the dashboard ⟶ system displays currently active alerts relevant to the user's location.

3.2.3 Functional Requirements

REQ-ALERT-1: The system shall allow an Authority or Administrator to create an alert containing disaster type, title, description, affected area, severity, date/time, safety instructions, and status.

REQ-ALERT-2: The system shall display active alerts to citizens whose registered or current location falls within the affected area.

REQ-ALERT-3: The system shall allow an alert's status to be updated (e.g., Active, Resolved) and reflect the change to users in near real time.

REQ-ALERT-4: The system shall maintain a history of past alerts for reference and reporting.

3.3 Incident Reporting

3.3.1 Description and Priority

Allows citizens to report incidents occurring around them and tracks each incident through its lifecycle. Priority: High.

3.3.2 Stimulus/Response Sequences

- Citizen submits an incident report with type, description, and location → system records the incident with status "Reported" and notifies relevant Authorities/NGOs.
- Authority reviews an incident → system updates its status through Under Review, Assigned, In Progress, to Resolved.

3.3.3 Functional Requirements

REQ-INC-1: The system shall allow a citizen to report an incident with type, description, location, time, and severity.

REQ-INC-2: The system shall allow an optional image/evidence attachment on an incident report, if implemented.

REQ-INC-3: The system shall track each incident through the statuses: Reported, Under Review, Assigned, In Progress, Resolved.

REQ-INC-4: The system shall notify the reporting citizen when the status of their incident report changes.

REQ-INC-5: The system shall allow Authorities and NGOs to view a list of incidents filtered by area, type, or status.

3.4 Emergency Help Requests

3.4.1 Description and Priority

The core module connecting affected people directly with responders. Allows citizens to request specific assistance and allows volunteers/NGOs to accept and fulfill requests. Priority: High.

3.4.2 Stimulus/Response Sequences

- Citizen creates a help request specifying type of help, location, description, and number of people affected → system records the request and makes it visible to nearby volunteers/NGOs.
- Volunteer/NGO accepts a request → system updates status from Pending to Accepted to In Progress to Completed, notifying the citizen at each change.

3.4.3 Functional Requirements

REQ-HELP-1: The system shall allow a citizen to create a help request specifying request type (Medical, Food, Water, Rescue, Shelter, Transportation, Clothing, Other), location, description, number of people, and priority.

REQ-HELP-2: The system shall display open help requests to volunteers and NGOs, sorted or filterable by proximity and priority.

REQ-HELP-3: The system shall allow a volunteer or NGO to accept a help request, after which it is no longer visible as "open" to other responders.

REQ-HELP-4: The system shall track each help request through the statuses: Pending, Accepted, In Progress, Completed.

REQ-HELP-5: The system shall notify the requesting citizen whenever the status of their help request changes.

REQ-HELP-6: The system shall record a timestamp for creation and for each status change of a help request, to support reporting and auditing.

3.5 Shelters and Essential Services

3.5.1 Description and Priority

Helps citizens discover nearby shelters, hospitals, police stations, and relief/distribution centers. Priority: Medium-High.

3.5.2 Stimulus/Response Sequences

- Citizen requests nearby services → system uses the citizen's location to return a distance-sorted list of relevant facilities.

3.5.3 Functional Requirements

REQ-SVC-1: The system shall maintain a directory of shelters and essential services with name, type, location, contact information, and availability/status.

REQ-SVC-2: The system shall allow a citizen to retrieve nearby services sorted by distance from their current or entered location.

REQ-SVC-3: The system shall allow Administrators and Authorities to add, update, or deactivate service entries.

REQ-SVC-4: The system shall allow filtering of services by type (e.g., Hospital, Shelter, Police Station, Relief Center).

3.6 NGO and Volunteer Coordination

3.6.1 Description and Priority

Provides NGOs and volunteers with a centralized, structured view of open help requests instead of relying on ad hoc discovery. Priority: High.

3.6.2 Stimulus/Response Sequences

- NGO views its coordination dashboard → system displays active requests, assigned volunteers, and available resources in one place.
- NGO assigns a volunteer to a request → system links the volunteer to that request and updates both parties' views.

3.6.3 Functional Requirements

REQ-COORD-1: The system shall provide NGOs with a dashboard listing active help requests, assigned volunteers, and resource status.

REQ-COORD-2: The system shall allow an NGO to assign a registered volunteer to a specific help request.

REQ-COORD-3: The system shall allow volunteers to view requests assigned to them and update their status (Accepted, In Progress, Completed).

REQ-COORD-4: The system shall prevent the same open request from being simultaneously accepted by more than one volunteer/NGO.

3.7 Resource Shortage Management

3.7.1 Description and Priority

Tracks relief-resource availability and shortages by area so that relief organizations can coordinate supply. Priority: Medium.

3.7.2 Stimulus/Response Sequences

- NGO reports resource quantities for an area → system records resource type, required quantity, available quantity, location, and priority.
- Another NGO views resource status for an area → system displays current shortages so supply can be coordinated without duplication.

3.7.3 Functional Requirements

REQ-RES-1: The system shall allow an NGO to record and update resource entries (type, required quantity, available quantity, location, priority, status).

REQ-RES-2: The system shall allow any NGO or Authority to view resource status by area, to identify shortages.

REQ-RES-3: The system shall flag a resource as "Critical" or "Low" when available quantity falls below a configurable threshold relative to required quantity.

REQ-RES-4: The system shall maintain a history of resource updates for a given area to support tracking over time.

3.8 Missing Person and Family Safety

3.8.1 Description and Priority

Provides a centralized, searchable database for reporting and locating missing persons during a disaster, in place of scattered social-media posts. Priority: High.

3.8.2 Stimulus/Response Sequences

- User reports a missing person with name, age, gender, last known location, and description → system records the report and makes it searchable.
- User searches by name, location, or identifying information → system returns matching missing-person records.

3.8.3 Functional Requirements

REQ-MISS-1: The system shall allow a user to report a missing person with name, age, gender, last known location, description, contact information, and optionally a photograph.

REQ-MISS-2: The system shall allow users to search missing-person records by name, location, or other identifying information.

REQ-MISS-3: The system shall allow the status of a missing-person report to be updated (e.g., Missing, Found) by the reporter or an authorized user.

REQ-MISS-4: The system shall protect missing-person contact details from being visible to unauthenticated users, per Section 5.3.

3.9 Relief Campaigns and Donations

3.9.1 Description and Priority

Allows verified NGOs to publish relief/donation campaigns while preventing unverified organizations from misrepresenting themselves. Priority: Medium.

3.9.2 Stimulus/Response Sequences

- Verified NGO creates a campaign with title, description, affected area, required resources, and donation information → system publishes the campaign to users.
- Unverified NGO attempts to create a campaign → system blocks the action until verification is complete.

3.9.3 Functional Requirements

REQ-CAMP-1: The system shall allow only NGOs with "Verified" status to create or publish relief campaigns.

REQ-CAMP-2: The system shall allow a campaign to include organization, title, description, affected area, required resources, donation information, and status.

REQ-CAMP-3: The system shall allow citizens to view active, verified relief campaigns.

REQ-CAMP-4: The system shall allow an administrator to suspend or remove a campaign that violates platform policy.

3.10 Notifications

3.10.1 Description and Priority

Keeps users informed of relevant events and status changes across all modules. Priority: Medium-High.

3.10.2 Stimulus/Response Sequences

- A tracked event occurs (e.g., help request accepted, alert issued for the user's area) → system generates and delivers a notification to the affected user.

3.10.3 Functional Requirements

REQ-NOTIF-1: The system shall notify a user when a disaster alert is issued for their registered or current area.

REQ-NOTIF-2: The system shall notify a citizen when their help request is accepted or its status changes.

REQ-NOTIF-3: The system shall notify a user when a missing-person report they filed receives an update.

REQ-NOTIF-4: The system shall allow users to view a history of their past notifications within the application.

3.11 Admin Management

3.11.1 Description and Priority

Provides the administrator with centralized oversight of users, NGOs, disasters, and reported content. Priority: High.

3.11.2 Stimulus/Response Sequences

- Administrator reviews a pending NGO registration → system allows approval or rejection, updating the NGO's verification status accordingly.
- Administrator reviews reported/flagged content → system allows the content to be removed or the associated account to be suspended.

3.11.3 Functional Requirements

REQ-ADMIN-1: The system shall allow an administrator to approve or reject NGO verification requests.

REQ-ADMIN-2: The system shall allow an administrator to view and manage all users, NGOs, disasters, alerts, incidents, help requests, missing-person reports, resources, and campaigns.

REQ-ADMIN-3: The system shall allow an administrator to suspend a user or NGO account found to be in violation of platform policy.

REQ-ADMIN-4: The system shall provide the administrator with summary activity reports (e.g., counts of open requests, active alerts) for monitoring system health.

4. External Interface Requirements

4.1 User Interfaces

DisasterConnect provides role-specific dashboards for each user class, described below. All interfaces must remain simple and low-friction, since citizen users may be operating the system under stress during an active disaster.

- Citizen Dashboard: active alerts, nearby services, request-help action, report-incident action, missing persons, relief information, notifications.
- Volunteer Dashboard: nearby help requests, my assigned requests, completed requests.
- NGO Dashboard: active requests, volunteers, resources, shortages, relief campaigns, reports.
- Admin Dashboard: users, NGOs, volunteers, disasters, alerts, incidents, help requests, missing persons, resources, campaigns, reports.

Standard elements such as navigation, a help/FAQ link, and consistent error-message styling shall appear across all dashboards. Detailed screen-level UI design is to be documented separately in a UI specification/mockups.

4.2 Hardware Interfaces

DisasterConnect does not require dedicated hardware. It relies on the standard networking and geolocation hardware present in users' smartphones, tablets, or computers (e.g., GPS/location services for nearby-service and nearby-request features).

4.3 Software Interfaces

- Database: a relational database (e.g., PostgreSQL/MySQL) storing users, disasters, alerts, incidents, help requests, NGOs, volunteers, resources, missing persons, and services.
- Backend Framework: a REST API application layer (e.g., Spring Boot), exposing endpoints consumed by the frontend.
- Frontend: a browser-based client application consuming the REST APIs over HTTPS.
- Mapping/Geolocation Service: an external mapping API may be used for distance calculation and location display (to be finalized in design).

4.4 Communications Interfaces

- All client-server communication shall use HTTPS.
- The system shall use standard JSON payloads over REST endpoints for all data exchange between frontend and backend.
- Email or push notifications may be used to deliver alerts and status-change notifications (mechanism to be finalized in design).
- Future versions may integrate SMS gateways for users without consistent internet access; this is out of scope for Version 1.0.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall respond to emergency help-request submissions within 2 seconds under normal load.
- The system shall be able to support a sharp increase in concurrent users during an active disaster event without significant degradation in response time for core emergency features (alerts, help requests, incident reporting).
- Search operations (nearby services, missing-person search) shall return results within 3 seconds under normal load.

5.2 Safety Requirements

- The system shall never block or delay the display of critical safety information (active alerts, safety instructions) behind non-essential features.
- The system shall clearly distinguish verified emergency/authority information from unverified user-submitted content to avoid misleading affected users.
- Where the system cannot guarantee a response (e.g., no volunteers available nearby), it shall clearly communicate this to the user rather than implying help is guaranteed.

5.3 Security Requirements

- The system shall require authentication for all functionality beyond viewing public alerts and public service listings.
- Passwords shall be stored using a strong one-way hashing algorithm; they shall never be stored or transmitted in plain text.
- The system shall enforce role-based authorization so that, for example, a citizen cannot access administrator or NGO-only functionality.
- Sensitive personal data (contact numbers, missing-person details) shall only be visible to authenticated users with a legitimate need (e.g., responders, administrators), not the general public.
- NGO accounts shall be verified by an administrator before being permitted to publish relief campaigns or appear as trusted responders.

5.4 Software Quality Attributes

- Usability: the interface shall be simple and low-friction, prioritizing ease of use over ease of learning, given that citizen users may be under stress.
- Availability: core emergency features shall remain available even during periods of high demand.
- Reliability: submitted help requests and incident reports shall not be lost once acknowledged by the system.
- Scalability: the backend shall be designed to scale horizontally to accommodate demand spikes during large-scale disasters.
- Maintainability: the backend shall follow a modular, layered architecture (presentation, business logic, data access) so individual components can be modified independently.

6. Other Requirements

- Database requirements: the schema shall include, at minimum, the entities described in Appendix B (User, Disaster, Incident, Help Request, NGO, Volunteer, Resource, Missing Person, Shelter/Service).
- Internationalization: Version 1.0 targets English-language content; the data model should not preclude future localization.
- Academic/legal requirements: as an individual academic project, the system is not intended for production deployment without further legal and compliance review (e.g., data-protection regulations applicable to personal and emergency data).

Appendix A: Glossary

Term	Definition
Affected Citizen	A general public user who may need assistance during a disaster.
Alert	A published notice informing users of a disaster affecting a given area.
Help Request	A citizen-submitted request for a specific type of emergency assistance.
Incident	A citizen-reported occurrence (e.g., flooding, structural collapse) requiring attention.
NGO	Non-Governmental Organization; a verified relief organization coordinating larger-scale response.
Missing Person Report	A record created to help locate a person separated from their family during a disaster.
Relief Campaign	A verified NGO-run initiative to gather resources or donations for a specific disaster.
Resource Shortage	A condition where the available quantity of a resource in an area is below what is required.
Verification (NGO)	The administrator-approved status confirming an NGO is a legitimate relief organization.
Volunteer	A registered individual who accepts and fulfills help requests on the platform.

Appendix B: Analysis Models

B.1 High-Level Architecture

DisasterConnect follows a layered architecture consisting of a frontend/user-interface layer, a REST API layer, a backend application layer (services, business logic, and validation), a repository layer, and the underlying database. Presentation, business logic, and data access are kept separate to support maintainability and independent evolution of each layer.

B.2 Core Entities and Relationships

The main entities are User, Disaster, Alert, Incident, Help Request, NGO, Volunteer, Resource, Missing Person, and Shelter/Service. A User can create many Help Requests, Incidents, and Notifications. A Disaster is associated with many Alerts and Incidents. An NGO is associated with many Volunteers, Resources, and Relief Campaigns. A Help Request is linked to the Volunteer or NGO that accepts it.

User

Attribute	Description
user_id	Unique identifier for the user.
name	Full name of the user.
email	Email address, used for login and notifications.
phone	Contact phone number.
password	Hashed password credential.
role	Citizen, Volunteer, NGO, Authority, or Administrator.
location	Registered or current location of the user.

Disaster

Attribute	Description
disaster_id	Unique identifier for the disaster event.
type	Flood, earthquake, cyclone, fire, landslide, or other.
title	Short title of the disaster event.
description	Detailed description of the event.
affected_area	Geographic area impacted.
severity	Severity level (e.g., Low, Medium, High).
status	Active, Resolved, etc.
timestamp	Date/time the disaster was recorded.

Incident

Attribute	Description
incident_id	Unique identifier for the incident report.
reporter_id	Reference to the reporting user.
type	Type of incident reported.
description	Description of the incident.
location	Location of the incident.
severity	Severity level of the incident.
status	Reported, Under Review, Assigned, In Progress, or Resolved.
timestamp	Date/time the incident was reported.

Help Request

Attribute	Description
request_id	Unique identifier for the help request.
user_id	Reference to the requesting citizen.
request_type	Medical, Food, Water, Rescue, Shelter, Transportation, Clothing, Other.
description	Details of the assistance needed.
location	Location where help is needed.
priority	Priority level of the request.
assigned_to	Reference to the volunteer/NGO handling the request.
status	Pending, Accepted, In Progress, or Completed.
timestamp	Date/time the request was created.

NGO

Attribute	Description
ngo_id	Unique identifier for the NGO.
name	Name of the organization.
contact	Contact information for the NGO.
verification_status	Pending or Verified.
location	Primary operating location of the NGO.

Volunteer

Attribute	Description
volunteer_id	Unique identifier for the volunteer.
user_id	Reference to the underlying user account.
availability	Current availability status.
location	Current or registered location.
skills	Relevant skills or capabilities.

Resource

Attribute	Description
resource_id	Unique identifier for the resource entry.
type	Type of resource (food, water, medicine, blankets, etc.).
quantity	Quantity available.
location	Location associated with the resource.
status	Available, Low, or Critical.

Missing Person

Attribute	Description
missing_id	Unique identifier for the missing-person report.
reported_by	Reference to the reporting user.
name	Name of the missing person.
age	Age of the missing person.
description	Physical description or identifying details.
last_seen_location	Last known location of the missing person.
status	Missing or Found.

Shelter / Service

Attribute	Description
service_id	Unique identifier for the service entry.
name	Name of the shelter or service facility.
type	Hospital, Shelter, Police Station, Fire Station, Relief Center, etc.
location	Location of the facility.

Attribute	Description
capacity_availability	Current capacity or availability status.
contact	Contact information for the facility.

B.3 Overall Workflow

A disaster triggers alert publication by an Authority/Administrator. Citizens may report incidents and raise help requests, both of which carry location and requirement details. NGOs and volunteers view open requests, accept them, and update their status through to completion. In parallel, NGOs track and report resource shortages, and citizens may report or search for missing family members. Verified NGOs may run relief campaigns to gather additional resources. All relevant status changes generate notifications to the affected users.

Appendix C: Issues List

- TBD: Final choice of technology stack (backend framework, database, mapping/geolocation provider).
- TBD: Whether SMS/offline-fallback notifications are in scope for Version 1.0 or deferred to a future release.
- TBD: Exact threshold values used to flag a resource as "Low" vs. "Critical" in Resource Shortage Management.
- TBD: Data-retention and deletion policy for missing-person and incident records after a disaster is resolved.
- Open: Process and criteria for administrator NGO verification are to be defined in detail during design.